

Homework #2: Implementing Machine Learning for Face Recognition

1 The Assignment in One Minute

In this assignment, you will experiment with a simple machine learning based face recognition system. Rather than building the entire system from scratch, you have already been given facial landmark data and starter Python code.

Your job is to investigate a simple question:

How can we change the classifier and the way we represent a face to improve recognition?

You will explore this question using two datasets: the Caltech Faces Dataset¹ and the Specs on Faces (SoF) Dataset².

Minimum requirements

For the **Caltech dataset**, you must:

- explore at least **two classifiers other than kNN**; and
- implement at least **one new feature representation** beyond the provided Euclidean distance representation.

For the **SoF dataset**, you must:

- explore at least **two additional classifiers that are different from the classifiers you used for Caltech**; and
- implement at least **one new feature representation that is different from the new representation you used for Caltech**.

Therefore, across the entire assignment, you will explore at least:

4 different classifiers + 2 different feature representations

2 What You Have Been Given

You have been provided with the following files:

¹<https://data.caltech.edu/records/6rjah-hdv18>

²<https://sites.google.com/view/sof-dataset>

1. `main.py`
2. `X-5-Caltech.npy`, `y-5-Caltech.npy`
3. `X-68-Caltech.npy`, `y-68-Caltech.npy`
4. `X-5-SoF.npy`, `y-5-SoF.npy`
5. `X-68-SoF.npy`, `y-68-SoF.npy`

You are **not** expected to collect images, detect faces, or extract facial landmarks yourself. That work has already been completed so that you can focus on machine learning experimentation.

2.1 `main.py`: Your Starting Code

The file `main.py` contains the code you will modify.

The starter code currently uses a k -Nearest Neighbors (kNN) classifier with the default parameters provided by Scikit-learn:³

```
clf = KNeighborsClassifier()
```

This is your **starting point**, not the classifier you are expected to use for the entire assignment.

Scikit-learn includes many other supervised learning algorithms that you can explore:⁴

You do not need to understand every classifier in Scikit-learn. Instead, use the documentation and what we have discussed in class to identify approaches that seem reasonable, try them, and examine what happens.

2.2 The `x` and `y` Files

The provided `.npy` files are NumPy arrays containing the data used by your classifier.

- The `X-...` files contain the (x, y) coordinates of facial landmarks.
 - `X-5-Caltech.npy` and `X-5-SoF.npy` contain **5 landmarks** per face. Their shape is:

```
[num_images, 5, 2]
```

³<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>

⁴https://scikit-learn.org/stable/supervised_learning.html#supervised-learning

- `X-68-Caltech.npy` and `X-68-SoF.npy` contain **68 landmarks** per face. Their shape is:

```
[num_images, 68, 2]
```

- The `y-...` files contain the corresponding identity labels. Each label tells us which person is represented by the corresponding entry in `X`.

In other words:

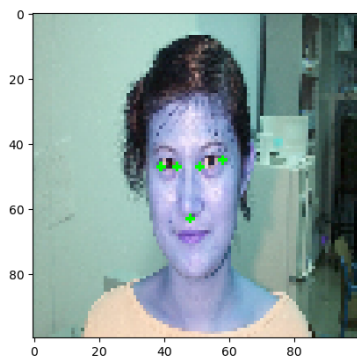
`X` = information describing the face `y` = identity of the face

3 Choosing 5 or 68 Landmarks

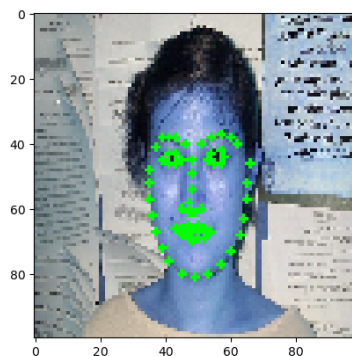
For each dataset, you may use:

- the 5-landmark version,
- the 68-landmark version, or
- both, if you want to compare them.

You are **not required** to test every possible dataset/landmark combination.



(a) 5 landmarks



(b) 68 landmarks

The two options provide different amounts of information about the face. Five landmarks give you a simpler representation, while 68 landmarks provide much more information about facial geometry.

Important: Changing from 5 landmarks to 68 landmarks does NOT count as your new feature representation.

The landmarks are the *raw information available to you*. A feature representation describes *how you transform those landmarks into values that the classifier will use*.

4 The Current Feature Representation

The classifier does not currently use the landmark coordinates directly.

Instead, the starter code computes the Euclidean distance between every pair of landmarks:

```
features = []

for k in range(num_identities):
    person_k = X[k]
    features_k = []

    for i in range(person_k.shape[0]):
        for j in range(person_k.shape[0]):

            p1 = person_k[i,:]
            p2 = person_k[j,:]

            features_k.append(
                math.sqrt(
                    (p1[0] - p2[0])**2 +
                    (p1[1] - p2[1])**2
                )
            )

    features.append(features_k)

features = np.array(features)
```

For example, if you use 5 landmarks, the code calculates:

$$5 \times 5 = 25$$

distance values for each face.

These 25 values become the **feature vector** provided to the classifier.

With 68 landmarks, the same approach produces:

$$68 \times 68 = 4624$$

features per face.

5 Your Feature Representation Experiment

For each dataset, you must explore at least **one way of representing the landmarks other than the provided all-pairs Euclidean distance approach**.

There is no single correct feature representation.

The question you should ask yourself is:

What information about facial geometry might help distinguish one person's face from another?

What Counts as a New Feature Representation?

A new feature representation changes what information you calculate from the landmarks.

For example:

Change	Counts as a new feature representation?
Change from 5 landmarks to 68 landmarks	No
Calculate facial distance ratios instead of only distances	Yes
Calculate angles between selected landmarks	Yes
Use a different classifier	No
Change k from 5 to 3 in kNN	No

6 Classifier Exploration

For each dataset, you will also explore classifiers available through Scikit-learn.

Your goal is not to randomly try every classifier you can find. Instead:

1. Read enough about a classifier to understand its basic idea.
2. Decide why it might make sense for your data.
3. Implement it.
4. Examine its accuracy.
5. Use the result to decide what to try next.

You may also explore the parameters of each classifier.

For example, if you are working with kNN, you might investigate how changing the number of neighbors affects performance.

However:

Changing classifier parameters does not create a new classifier.

For example:

```
KNeighborsClassifier(n_neighbors=5)
KNeighborsClassifier(n_neighbors=3)
KNeighborsClassifier(n_neighbors=1)
```

are three versions of **the same classifier**. They do not count as three different classifiers.

7 A Suggested Way to Approach the Assignment

You are free to organize your experiments differently, but if you are unsure where to begin, the following workflow is a good starting point.

Step 1: Run the starter code

Run `main.py` without changing anything.

Record the baseline accuracy.

Ask yourself:

- How well does the current system work?
- What classifier is being used?
- What features are being used?
- Am I using 5 landmarks or 68 landmarks?

Step 2: Explore your first new classifier

Look through the Scikit-learn documentation and select a classifier that seems reasonable.

Do not worry about finding the “perfect” classifier.

Try it and record what happens.

Step 3: Explore another classifier

Choose a second classifier.

Again, think about *why* it may be appropriate and compare its performance with what you have already tested.

Step 4: Experiment with a new feature representation

Now change the way the landmark coordinates are transformed into features.

Compare the new results with your earlier results.

Step 5: Make additional changes if they help you answer a question

At this point, you may want to:

- tune classifier parameters,
- compare 5 versus 68 landmarks,
- combine feature ideas,
- revisit a classifier that previously looked promising, or
- test another reasonable idea.

You are encouraged to experiment, but you are not expected to try every possible combination.

8 Caltech and SoF Must Be Different

You will repeat the general experimentation process for both datasets, but the required experiments must differ.

Specifically:

- The two additional classifiers you count toward the **Caltech requirement** cannot also count toward the **SoF requirement**.
- The new feature representation you count toward the **Caltech requirement** must be different from the new feature representation you count toward the **SoF requirement**.

9 What Are You Trying to Optimize?

Accuracy is important because it helps you compare your experiments. However, the goal of this assignment is not simply to obtain the highest possible accuracy.

A useful experiment can also produce worse accuracy.

For example, suppose you try a feature representation and accuracy decreases substantially. That result still teaches you something: the information represented by those features may not be useful for distinguishing identities, or the classifier you selected may not work well with that representation.

What matters is that you can explain:

- what you changed,
- why you changed it,
- what happened, and
- what you learned from the result.

You will need to discuss your experiments in your report, so record your results as you work.

10 Report

Prepare a concise, professional report summarizing your experiments.

Formatting Requirements

The report must:

1. be typed and professionally formatted;
2. be no longer than **2 pages**;
3. use **11-point Times New Roman**;
4. use **1.15 line spacing**; and
5. use **1-inch margins**.

The 2-page limit means you should focus on your most important experiments and findings rather than describing every line of code.

Required Report Sections

1. Classifier Exploration

For both datasets:

- identify the classifiers you tested;
- briefly explain why you selected them;
- identify important parameter settings;
- report and compare their performance; and
- explain what you learned from the comparison.

A table is encouraged if it helps you summarize several experiments efficiently.

2. Feature Representation

Explain the additional feature representation used for each dataset.

For each representation, explain:

- what you calculated from the landmarks;
- what information about the face you hoped the features would capture; and
- how the new representation affected your results.

Remember that the new feature representation must be different for Caltech and SoF.

11 What to Submit

Submit **one .zip file** containing:

1. your report as a .pdf; and
2. your modified .py file(s).

Include the code needed to reproduce the results discussed in your report.

You do **not** need to submit intermediate code versions or every experimental draft.

12 How You Will Be Graded

Your grade is based primarily on whether you complete the required exploration and whether you can clearly explain your decisions and findings.

Caltech Dataset

You should:

1. explore at least two unique classifiers beyond kNN;
2. explore at least one new feature representation beyond Euclidean distance;
3. reasonably explain what the classifiers do, why you selected them, and how they performed;
and
4. reasonably explain what your new feature representation was intended to capture and how it affected your results.

SoF Dataset

You should:

1. explore at least two unique classifiers beyond kNN that are different from your Caltech classifier choices;
2. explore at least one new feature representation that is different from your Caltech feature representation;
3. reasonably explain what the classifiers do, why you selected them, and how they performed;
and
4. reasonably explain what your new feature representation was intended to capture and how it affected your results.

What does “reasonably explain” mean?

You do not need to provide a mathematical derivation of every classifier.

Your explanation should simply provide enough information for the reader to understand:

- why you chose the approach,
- what you changed,
- what happened, and
- what you learned from the result.

The focus is on **thoughtful experimentation**, not on finding the “perfect” model.

13 Common Questions

Do I have to use both 5 and 68 landmarks?

No. You may use 5 landmarks, 68 landmarks, or both. Explain your choice when it is relevant to your experiments.

Does switching from 5 landmarks to 68 landmarks count as my new feature representation?

No. You are changing the amount of landmark information available, but you have not changed how those landmarks are converted into features.

Does changing k in kNN count as trying another classifier?

No. You are tuning a parameter of the same classifier.

Can I tune classifier parameters?

Yes. Parameter exploration is encouraged. It simply does not replace the requirement to explore different classifiers.

Can I use the same two classifiers for Caltech and SoF?

Not for the minimum assignment requirement. You must explore at least four different classifiers across the two datasets.

Can I use the same new feature representation for Caltech and SoF?

No. The feature representation used to satisfy the Caltech requirement must differ from the one used to satisfy the SoF requirement.

Do I need to try every classifier or feature representation?

Absolutely not. Make thoughtful choices, test them, and learn from the results.

What if one of my experiments makes the accuracy worse?

That is okay. Not every experiment will improve performance. A well-motivated experiment that produces a worse result can still teach you something useful. Explain what you tried and what you learned.

Am I graded on obtaining the highest accuracy?

No. Accuracy helps you evaluate and compare your approaches, but grading focuses on the quality of your exploration, comparisons, and explanations.